Echelon forms

A matrix is said to be in the **row echelon form** if it satisfies the following properties

- A row with only zero entries should be placed below all rows with some non-zero entry.
- The first non-zero entry in a row should be to the right of the first non-zero entry in the row above it.

The leading non-zero entry in each row in the row echelon form is called a **pivot** element. A matrix is said to be in the **reduced row echelon form** if it is in the row echelon form and in addition satisfies the following

- Each pivot is 1.
- The column containing a pivot has zero entries everywhere else in that column.

For example, the first two matrices are in reduced row-echelon form and the last two are in row echelon form

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad B = \begin{pmatrix} 0 & 1 & 5 & 0 & 6 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}; \quad C = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 4 & 5 \\ 0 & 0 & 3 & 7 \end{pmatrix}; \quad D = \begin{pmatrix} 1 & 1 & 1 & 4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

The reason is, for example in the matrix B, the pivot of first row is in 2nd column and every other entry in the 2nd column is zero. The pivot of the second row is in the 4th column and every other entry in the 4th column is zero. The third row does not contain any pivot. C is not in reduced row-echelon form as the pivot of second row is not equal to 1. The following matrix is also not in reduced row echelon form

$$E = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 4 & 2 \end{pmatrix}$$

as the pivot of the second row is in 2nd column and is equal to 1. But the other entries in the 2nd column are not zero.

Note 1. As you might have observed, in the Gaussian elimination method, we reduce the augmented matrix to the row-echelon form, while in the Gauss-Jordan method, we reduce the augmented matrix to the reduced row-echelon form. We will see the application of echelon forms in Module 2 when we discuss basis and dimension. In Module 1, we will use echelon forms in solving system of equations by Gauss elimination method or Gauss-Jordan method.

Example 1. Transform the following matrix first into row echelon form and then into row-reduced echelon form.

$$A = \begin{pmatrix} 1 & -1 & -2 \\ 2 & -1 & -1 \\ 2 & 1 & 4 \end{pmatrix}$$

The steps are as follows

$$\begin{pmatrix} 1 & -1 & -2 \\ 2 & -1 & -1 \\ 2 & 1 & 4 \end{pmatrix} \xrightarrow[R_2:R_2-2R_1]{R_3:R_3-2R_1} \begin{pmatrix} 1 & -1 & -2 \\ 0 & -1 & -3 \\ 0 & -3 & -8 \end{pmatrix} \xrightarrow{R_3:R_3-3R_2} \begin{pmatrix} 1 & -1 & -2 \\ 0 & -1 & -3 \\ 0 & 0 & 1 \end{pmatrix}$$

We have obtained the row echelon form. The remaining steps are as follows.

$$\begin{pmatrix} 1 & -1 & -2 \\ 0 & -1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2:(-1)R_2} \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1:R_1+R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_1:R_1-R_3]{R_2:R_2-3R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Example 2. Transform the following matrix first into row echelon form and then to reduced row echelon form

$$\begin{pmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{pmatrix}$$

Only the intermediary matrices are given. Identify the corresponding row operations.

$$\begin{pmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}, \qquad \text{row echelon form}$$

$$\rightsquigarrow \begin{pmatrix} 3 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{pmatrix}, \qquad \text{reduced row echelon form.}$$

Example 3. Solve the linear system by Gauss-Jordan method

$$\begin{aligned} x - y - 2z + t &= 0 \\ 2x - y - 3z + 2t &= -6 \\ -x + 2y + z + 3t &= 0 \\ x + y - z + 2t &= 1 \end{aligned}$$

We form the augmented matrix and since the method is Gauss-Jordan, we transform the coefficient part to row-reduced echelon form.

$$\left(\begin{array}{cccc|c} 1 & -1 & -2 & 1 & 0 \\ 2 & -1 & -3 & 2 & -6 \\ -1 & 2 & 1 & 3 & 0 \\ 1 & 1 & -1 & 2 & 1 \end{array}\right) \rightarrow \left(\begin{array}{cccc|c} 1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 & -6 \\ 0 & 1 & -1 & 4 & 2 \\ 0 & 2 & 1 & 1 & 1 \end{array}\right)$$

using the transformations $R_2 : R_2 - R_1$, $R_2 : R_3 + R_1$ and $R_4 : R_4 - R_1$,

$$\left(\begin{array}{cccc|c} 1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 & -6 \\ 0 & 1 & -1 & 4 & 2 \\ 0 & 2 & 1 & 1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & -1 & 1 & -6 \\ 0 & 1 & 1 & 0 & -6 \\ 0 & 0 & -2 & 4 & 8 \\ 0 & 0 & -1 & 1 & 13 \end{array}\right)$$

using the transformations $R_1 : R_1 + R_2$, $R_3 : R_3 - R_2$ and $R_4 : R_4 - 2R_2$. Observe that all the entries in the third row is divisible by 2. We use this fact to obtain the pivot as 1 in the third row

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 1 & -6 \\ 0 & 1 & 1 & 0 & -6 \\ 0 & 0 & 1 & -2 & -4 \\ 0 & 0 & -1 & 1 & 13 \end{array}\right)$$

Now transform

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 1 & -6 \\ 0 & 1 & 1 & 0 & -6 \\ 0 & 0 & 1 & -2 & -4 \\ 0 & 0 & -1 & 1 & 13 \end{array}\right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -10 \\ 0 & 1 & 0 & 2 & -2 \\ 0 & 0 & 1 & -2 & -4 \\ 0 & 0 & 0 & -1 & 9 \end{array}\right)$$

using the transformations $R_1 : R_1 + R_3$, $R_2 : R_2 - R_3$ and $R_4 : R_4 + R_3$. The pivot of the fourth row is made 1 by dividing by -1

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -10 \\ 0 & 1 & 0 & 2 & -2 \\ 0 & 0 & 1 & -2 & -4 \\ 0 & 0 & 0 & 1 & -9 \end{array}\right)$$

Finally, we obtain the row reduced echelon form

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -10 \\ 0 & 1 & 0 & 2 & -2 \\ 0 & 0 & 1 & -2 & -4 \\ 0 & 0 & 0 & 1 & -9 \end{array}\right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -19 \\ 0 & 1 & 0 & 0 & 16 \\ 0 & 0 & 1 & 0 & -22 \\ 0 & 0 & 0 & 1 & -9 \end{array}\right)$$

from which we obtain the solution as $t = -9$, $z = -22$, $y = 16$ and $x = -19$.

Example 4. Solve the following system of equation via Gauss elimination method

$$\begin{aligned} 3x - y + z + 2t &= -2 \\ x + 2y - z + t &= 1 \\ -x - 3y + 2z - 4t &= -6 \end{aligned}$$

We write the augmented matrix and transform to row echelon form as the method is Gauss elimination.

$$\left(\begin{array}{cccc|c} 3 & -1 & 1 & 2 & -2 \\ 1 & 2 & -1 & 1 & 1 \\ -1 & -3 & 2 & -4 & -6 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 1 \\ 0 & 1 & -1 & 3 & 5 \\ 0 & 0 & 1 & -20/3 & 10 \end{array}\right)$$

The coefficient part of the augmented matrix is row echelon form and hence we stop. The corresponding equations are

$$z - (20/3)t = -10, \qquad y - z + 3t = 5, \qquad x + 2y - z + t = 1$$

Note that we have not obtained a perfect triangular form and also the number of variables is less than the number of equations. Hence we will have an infinite number of solutions.

Let t be the free variable. Then the solutions are

$$\begin{aligned} z &= -10 + (20/3)t \\ y &= 5 + z - 3t = 5 - 10 + (20/3)t - 3t = -5 + (11/3)t \\ x &= 1 - 2y + z - t = 1 - (5/3)t \end{aligned}$$

The solution set is

$$\begin{aligned} S &= \left\{ \left(1 - \frac{5}{2}t, -5 + \frac{11}{3}t, -10 + \frac{20}{3}t, t\right) | t \in \mathbb{R} \right\} \\ &= \left\{ (1, -5, -10, 0) + t\left(-\frac{5}{2}, \frac{11}{3}, \frac{20}{3}, 1\right) | t \in \mathbb{R} \right\} \end{aligned}$$

Exercises

1. Solve, if possible, the following system of equation

$$\begin{aligned} x + y + z &= 4 \\ 3x - y - z &= 2 \\ x + 3y + 3z &= 8 \end{aligned}$$

2. Determine when the following augmented matrix represents a system of equation with an unique solution, infinite number of solutions and no solution.

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & a \\ 2 & 1 & 5 & b \\ 1 & -1 & 1 & c \end{array}\right)$$